# Tong Chen

470-641-4428 · txc5603@mavs.uta.edu
tongchen2010.github.io · github.com/tongchen2010 · linkedin.com/in/tong-chen-850935124

## Research Interests

Machine learning and AI for medical imaging, with a focus on Alzheimer's disease (AD) and Lewy body dementia (LBD): self-supervised representation learning for longitudinal brain MRI and disease-progression modeling across the cognitive impairment continuum.

## Education

**Ph.D. in Computer Science, The University of Texas at Arlington** Aug 2023 – Present
*Advisor: Dr. Dajiang Zhu (expected Dec 2026)*

**M.S. in Computer Science, Kennesaw State University** Jan 2021 – Dec 2022

**B.S. in Computer Science, Kennesaw State University** Aug 2012 – May 2018

## Selected Publications

(**Bold** = author; * = equal contribution. Full list at tongchen2010.github.io.)

1. Minheng Chen, **Tong Chen**, Yan Zhuang, et al. "Community-Level Modeling of Gyral Folding Patterns for Robust and Anatomically Informed Individualized Brain Mapping." *NeuroImage*, 2026.
2. **Tong Chen**, Minheng Chen, Jing Zhang, et al. "HiLoGraph: Hierarchical- Longitudinal Brain Network Representation Learning." *MICCAI*, 2026. **(Early accept, top 9%)**
3. **Tong Chen**, Minheng Chen, Jing Zhang, et al. "A Unified Continuous Staging Framework for Alzheimer's Disease and Lewy Body Dementia via Hierarchical Anatomical Features." *MICCAI*, 2025. **(Early accept, top 9%)**
4. Chao Cao*, **Tong Chen***, Nan Zhao*, et al. "Gyral-Sulcal-Net: An Integrated Network Representation of Brain Folding Patterns." *IEEE ISBI*, 2026. **(Co-first author)**

## Research Experience

**Graduate Research Assistant, UT Arlington (Advisor: Dr. Dajiang Zhu)** Aug 2023 – Present

- Developed graph neural network methods for multi-class dementia classification and continuous disease-progression modeling on large multi-cohort neuroimaging data.
- Built sMRI and DTI preprocessing and analysis pipelines for multi-cohort studies; first-author work at MICCAI and AAIC and a co-authored *NeuroImage* article.

**Research Assistant, UNC Chapel Hill (Advisor: Dr. Gang Li)** May 2025 – Aug 2025

- Developed cortical surface-based machine learning methods on Human Connectome Project (HCP) data.

## Technical Skills

**Programming:** Python, MATLAB **Frameworks:** PyTorch, PyTorch Geometric, scikit-learn
**Models:** Graph Neural Networks, Transformers, State-Space Models (Mamba), Multimodal LLMs
**Neuroimaging:** FreeSurfer, FSL, SPM12, CAT12, Connectome Workbench, DSI Studio, MRtrix3
**Datasets:** ADNI, NACC, AABC, HCP, BCP, in-house LBD cohorts